# CS 222: Computer Organization and Architecture

## End Sem Exam      Max Marks: 60, Max Time: 180 minutes

**(Answer precisely and mention explicitly all the steps. Write you assumption if not mentioned implicitly nor explicitly in the question. No credit will be given if answering without showing the work)**

1. Draw a flowchart to explain how the interrupt driven data transfer occurs. [3]

2. Let a processor architecture has not considered the handling of multiple interrupt requests. Without changing the hardware configuration how can you incorporate this feature? [3]

3. Assume a processor having 2-way set-associative TLB with 32 total entries and an LRU replacement policy. Physical addresses of 24 bits, Virtual addresses of 32 bits, Byte addressable memory. Page size is 64 KB. The following table gives the state of the TLB, providing the address translations for the relevant virtual addresses (using hexadecimal notation).

| TLB | | | | | | | |
|---|---|---|---|---|---|---|---|
| Index | Tag | Physical page number | Valid | Index | Tag | Physical page number | Valid |
| F | FFF | EF | 1 | 7 | 560 | 6F | 1 |
| | FFC | 71 | 1 | | 8F1 | A9 | 1 |
| E | FF9 | 28 | 1 | 6 | A31 | 31 | 1 |
| | FF4 | AB | 1 | | 3B0 | 49 | 1 |
| D | F55 | CD | 1 | 5 | 9E1 | 55 | 1 |
| | FF6 | 34 | 1 | | 041 | 00 | 1 |
| C | 446 | 1F | 1 | 4 | 151 | 48 | 1 |
| | 00A | BB | 0 | | 1D0 | 32 | 1 |
| B | 6AC | D5 | 1 | 3 | 0A0 | 6A | 1 |
| | 715 | 8E | 1 | | 9D5 | 56 | 1 |
| A | 31A | D4 | 1 | 2 | 68E | 60 | 1 |
| | 51D | 64 | 1 | | E71 | 78 | 1 |
| 9 | 39E | 5A | 1 | 1 | A34 | EE | 1 |
| | 43B | F8 | 1 | | 8BB | AA | 1 |
| 8 | 732 | DE | 1 | 0 | 7A0 | 99 | 0 |
| | 48F | B2 | 0 | | 3F0 | 94 | 1 |

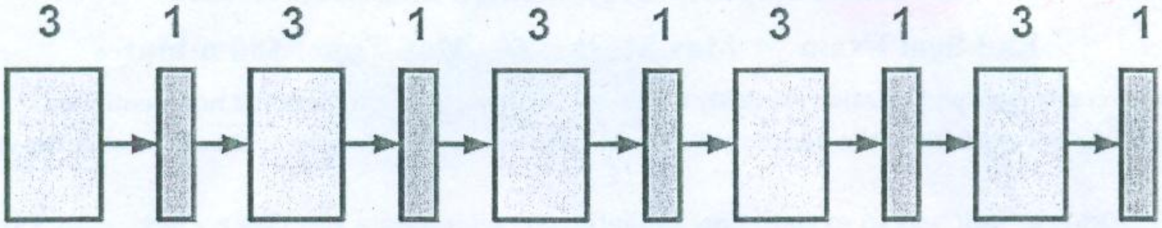
Write the physical address for the given virtual address (in hexadecimal) below.[6]
(i) FFFFABCD (iii) 446C CEBA (III) 446C CDEF (IV)7A00 2456

4. For programmed I/O, to increase efficiency, the I/O software could be written so that the processor periodically checks the status of the device. If the device is not ready, the processor can jump to other tasks. After some timed interval, the processor comes back to check status again. Consider the above scheme for outputting data one character at a time to a printer that operates at 10 characters per second (cps).What will happen if its status is scanned at every 200 ms? What will happen if its scanned period is reduced to every 75ms? [4]

5. Write a sequence of micro instructions in symbolic form (RTL) for instruction SUB R1, X, which subtracts the contents of location X. Assume that the processor has a logical unit and an adder (so direct subtraction operation is not possible). [6]

6. A 5 stage pipelined processor (pictured below) has maximum throughput of N instructions/second. Each stage requires 3 units of time for the combinational circuit and 1 unit for the pipeline registers. Assume that we combine all the combinational circuits into one big combinational circuit. What is the throughput of the resulting sequential implementation in terms of N. [4]

P.T.O




7. There is a data hazard in the following assembly code. Briefly describe the hazard (Showing in the pipeline sequence diagram), and rewrite the code with just enough NOPS inserted to eliminate the problem (no extra NOPs!). You should assume the processor is based on the 5 stage pipeline we discussed in class (stages Fetch, Decode, Execute, Memory, Writeback). [6]

   I1.: MOV DX, 3
   I2: MOV AX,4
   I3: ADD DX,AX

8. Identify all the hazards (data hazards and control hazards) in the following code and state whether the each hazard can be avoided by forwarding or whether some stall cycles are necessary (indicate the number of stall cycles necessary). Assume the 5 stage pipeline we looked at, with stages Fetch, Decode, Execute, Memory and Write-back. [6]

   I1. MOV AX,10
   I2. ADD AX,BX
   I3. MOV CX, [AX]
   I4. JMP I10

9. In a system with 4GB capacity memory, assume a task is divided into four equal-sized segments and that the system builds an eight-entry page descriptor table for each segment. Thus, the system has a combination of segmentation and paging. Assume also that the page size is 2 KBytes. [6]

   a. What is the maximum size of each segment?

   b. What is the maximum logical address space for the task?

   c. Assume that an element in physical location 00021ABC is accessed by this task. What is the format of the logical address that the task generates for it (show through a diagram)? What is the maximum physical address space for the system?

10. Demonstrate (with an example), an approach to implement LRU algorithm in a system which has 4-line cache (4 more marks if you use the minimum number of bits, total bits =6). [4+4=8]

**11.** Assume the following code address of the array A is 0, and other variables i,j,tot and zcnt are in registers (the only memory accesses are to A). What is the miss rate for the code assuming the cache holds 1024 bytes of data, is direct mapped and has a block size of 8 bytes? State ~~and~~ two different approaches to increase the hit rate by 50%. (if earlier it is 60 % it becomes 90%) . [2+3+3=[8]]

```
int A[256][2];
int i,j;
int tot=0;
int zcnt=0;
for (i=0;i<256;i++) {
for (j=0;j<2;j++) {
tot += A[i][j];
}
}
for (i=0;i<256;i++) {
for (j=0;j<2;j++) {
if (A[i][j]==0)
zcnt++;
}
}
```

// 4bytes integer //